EEG-Based Universal Game Controller Using Brain–Computer Interface

Project Team

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Chapter 1

Introduction

This document specifies the requirements for the **EEG-Based Universal Game Controller Using Brain–Computer Interface** , a brain-computer interface (BCI) application that enables users to control game actions using brain signals. The system is designed to decode motor imagery and eye blinks from EEG signals, along with data from the built-in gyroscope of the EEG headset, to generate game control commands. The readers of this document include developers responsible for building the system, project managers overseeing the development process, testers validating performance, and researchers or users interested in BCI-driven interaction design [8].

1.1 Problem Statement

Traditional methods of interacting with computer games rely heavily on physical input devices such as keyboards, mice, and handheld controllers. Although these devices are effective, they create accessibility barriers for individuals with motor impairments and limit the immersive potential of gaming experiences. Current EEG-based gaming solutions, such as those provided by commercial vendors such as Emotiv or Muse, are largely confined to proprietary platforms. Moreover, existing research prototypes suffer from issues such as lengthy calibration times, poor generalization between users. As a result, players are unable to use brain signals to interact with games in real-time. This system is being developed to overcome these challenges by creating an interface that translates EEG-based mental states and gyroscope data into standard gaming inputs. In doing so, the software addresses accessibility needs, provides a novel immersive input method for gamers, and offers researchers a flexible tool for exploring BCI applications beyond specialized test environments. The proposed solution addresses directly the issues of ecosystem lock-in, limited command sets, and inadequate real-world usability, ultimately enabling a reliable brain-controlled interface for modern gaming.

1.2 Scope

The scope of this project is to design and implement a brain–computer interface (BCI) system that allows users to control basic actions in the game Tuxemon using only EEG signals and gyroscope input. The system is intentionally limited to two distinct commands which will be mapped to two in-game actions. This keeps the project within a manageable range while still demonstrating the practicality of EEG-based control in gaming. In addition to EEG, gyroscope data will be incorporated to capture simple head movements, thereby enhancing control without adding excessive complexity. The solution will focus on real-time acquisition of brain signals through Smarting EEG devices, preprocessing of the data to remove noise, and classification of EEG patterns into meaningful control commands.Integration with the game Tuxemon will be achieved by mapping EEG and gyro signals to predefined keyboard inputs so that users can interact with the game environment seamlessly. The proposed project does not aim to cover a wide range of EEG commands or test across multiple games; instead, it establishes a proof-of-concept that EEG-driven gaming is both feasible and enjoyable. The boundaries of the system exclude medical-grade diagnostics, multi-class EEG recognition, or integration with complex commercial gaming platforms. The project is also limited to single-user interaction and does not involve multi-player EEG synchronization. Its focus is strictly on demonstrating functionality and usability within a controlled experimental setup. By narrowing the scope to only two commands and one test game, the project ensures achievable goals, measurable outcomes, and a clear path to evaluation. Ultimately, the system serves as a foundation for future extensions where additional commands, more sophisticated models, and broader game support can be introduced.

1.3 Modules

The proposed project is divided into four modules, each addressing a distinct part of the system workflow. These modules ensure the project is structured from signal capture to final gameplay evaluation, covering acquisition, processing, integration, and performance testing.

1.3.1 Module 1: Signal Acquisition

This module establishes a stable connection with the Smarting EEG headset and gyroscope to capture real-time data streams.

1. Acquire EEG signals and head movement data in real time.

2. Synchronize EEG and gyroscope channels for unified processing.

1.3.2 Module 2: Signal Processing and Classification

Transforms raw EEG and gyro data into usable control commands through filtering and classification.

1. Apply noise filtering and artifact removal.
2. Classify two mental commands using EEGNet or SVM.
3. Interpret gyroscope data for head movement detection.

1.3.3 Module 3: Game Integration

Connects the classified EEG and gyro outputs to the selected game (Tuxemon) by mapping them to keyboard actions.

1. Map EEG-based commands to keypress actions (e.g., select, confirm).
2. Use gyro data for camera or directional movement.

1.3.4 Module 4: Evaluation and Testing

Assesses the system's performance, usability, and reliability.

1. Measure EEG classification accuracy and system response time.
2. Conduct user trials to evaluate accessibility and immersion.
3. Generate reports and visualization for performance metrics.

1.4 User Classes and Characteristics

User Class	Description
Gamers	Gamers represent one of the primary stakeholders of this system. This group includes casual and professional players who seek immersive and innovative gameplay experiences. The EEG-based control system offers them a new way to interact with games using brain signals, blinks, and head movement—eliminating the need for traditional controllers. It enhances engagement and introduces a futuristic, thought-controlled gaming experience while maintaining responsiveness and accuracy.
People with Disabilities	This group includes individuals with limited motor abilities who face challenges using standard input devices such as keyboards, mice, or controllers. The EEG-based Game Control System provides an accessible alternative that enables them to play games and interact with digital environments through neural and minimal physical cues. This inclusion-driven approach promotes independence, accessibility, and equal participation in gaming and digital entertainment.
Developer	Responsible for implementing the signal acquisition pipeline, deep learning model integration, and game interface communication. Should understand EEG signal structures and programming interfaces for hardware integration.
Tester	Evaluates system accuracy, latency, and robustness under different environmental and signal conditions. May also use the system to study human-computer interaction or neurogaming.
Project Manager	Oversees progress, ensures milestones are met, and maintains coordination between technical and usability teams. Tracks performance metrics like classification accuracy, latency, and calibration time.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-Case Representation

Since the proposed system is an interactive, user-driven application, the **Use Case** technique is most appropriate for describing its behavior. The following use case diagram (Figure 2.1) illustrates the interaction between the user and the main components of the system.

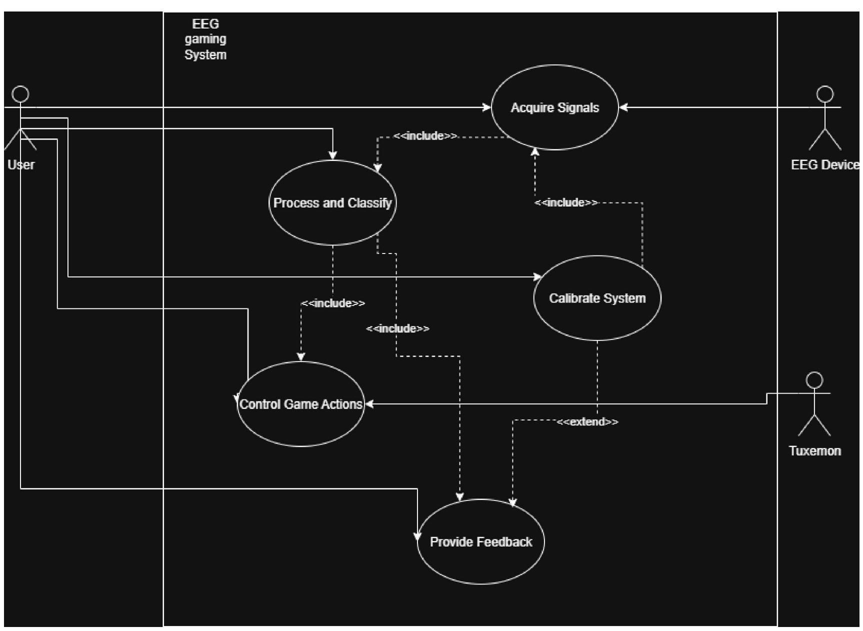

Figure 2.1: Use Case Diagram for EEG-Based Game Control System (MindPlay)

Primary Use Cases

- **UC1: Calibrate System —** The user performs short guided tasks (rest, hand imagery, blink) to train the model for personalized control.
- **UC2: Acquire Signals —** EEG and gyroscope data are streamed in real time from the headset.
- **UC3: Process and Classify Signals —** The system filters, preprocesses, and classifies neural activity into control commands.
- **UC4: Control Game Actions —** Classified commands are mapped to in-game actions such as move, jump, or select.
- **UC5: Provide Feedback —** The system gives real-time feedback (visual/audio) on detected commands or signal quality.

Detailed Use Cases

Table 2.1: Detailed Use Case UC1 – Calibrate System

Use Case: UC1 – Calibrate System	
Actors	User (Gamer or Person with Disability)
Precondition	EEG headset is connected and recognized by the system.
Postcondition	System collects sufficient labeled EEG and gyro data for personalized calibration.
Main Flow	1. The system prompts the user to begin calibration. 2. The user performs guided tasks such as "Rest," "Hand Imagery," and "Blink." 3. EEG and gyroscope data are recorded and labeled for each task. 4. The system fine-tunes the pretrained EEGNet model using new calibration data. 5. A performance summary is displayed after completion.
Alternate Flow	1. If EEG signal quality drops below a threshold, the system pauses calibration and asks the user to adjust the headset.
Exceptions	• Calibration timeout if data is not collected within 15 minutes.

Table 2.2: Detailed Use Case UC2 – Acquire Signals

Use Case: UC2 – Acquire Signals	
Actors	BCI System
Precondition	Headset is connected and calibration completed.
Postcondition	Continuous EEG and gyroscope streams are available for processing.

Main Flow	1. The system initializes data acquisition from the EEG headset. 2. Data streams are synchronized with timestamps. 3. The system buffers real-time data in short sliding windows (1–2 seconds). 4. Buffered data are passed to the preprocessing module.
Alternate Flow	1. If connection drops, the system attempts auto-reconnection.
Exceptions	• Sensor disconnection or signal noise beyond acceptable limits triggers a warning message.

Table 2.3: Detailed Use Case: UC3 – Process and Classify Signals

Use Case: UC3 – Process and Classify Signals	
Actors	BCI System
Precondition	Real-time EEG and gyro data are being received.
Postcondition	Classified commands (Rest, Hand Imagery, Blink) are generated for game mapping.

Main Flow	1. The system filters EEG signals (8–30 Hz) to remove noise and artifacts. 2. Eye blinks are detected and either classified or filtered based on amplitude thresholds. 3. The EEGNet model processes filtered data and outputs class probabilities. 4. Gyroscope data are processed to calculate head orientation or tilt. 5. The system fuses EEG and gyro information to produce a final command output.
Alternate Flow	1. If classifier confidence is below threshold, the system maintains the previous state ("Rest").
Exceptions	• Missing EEG or gyro data for more than 2 seconds pauses classification until restored.

Table 2.4: Detailed Use Case: UC4 – Control Game Actions

Use Case: UC4 – Control Game Actions	
Actors	User, Tuxemon
Precondition	Classifier outputs control commands in real time.
Postcondition	Game responds to EEG/gyro-based input commands.

Main Flow	1. Classified outputs are mapped to predefined game controls. 2. “Hand Imagery” triggers movement or action in the game. 3. “Blink” acts as a select or confirm command. 4. Gyroscope tilt controls direction or camera movement. 5. Commands are sent as keyboard/mouse events to the game.
Alternate Flow	1. If classification confidence drops below threshold, the system holds input until stable.
Exceptions	• If the game is not responding, the system queues commands and retries once connection resumes.

Table 2.5: Detailed Use Case: UC5 – Provide Feedback

Use Case: UC5 – Provide Feedback	
Actors	User
Precondition	System is actively processing EEG/gyro data.
Postcondition	User receives visual and/or audio feedback on performance and connection quality.

Main Flow	1. The system continuously monitors signal quality and classifier accuracy. 2. Feedback is displayed via visual indicators (bars, icons, or probability values). 3. The user can adjust headset placement if poor signal quality is detected. 4. At session end, a performance summary (accuracy, latency, false positives) is shown.
Alternate Flow	1. In minimal-UI mode, feedback is provided as audio tones or LED indicators.
Exceptions	• If headset is disconnected, system alerts the user and pauses feedback until reconnection.

Use Case UC1 – Calibrate System

Use Case UC2 – Acquire Signals

Use Case UC3 – Process and Classify Signals

Use Case UC4 – Control Game Actions

Use Case UC5 – Provide Feedback

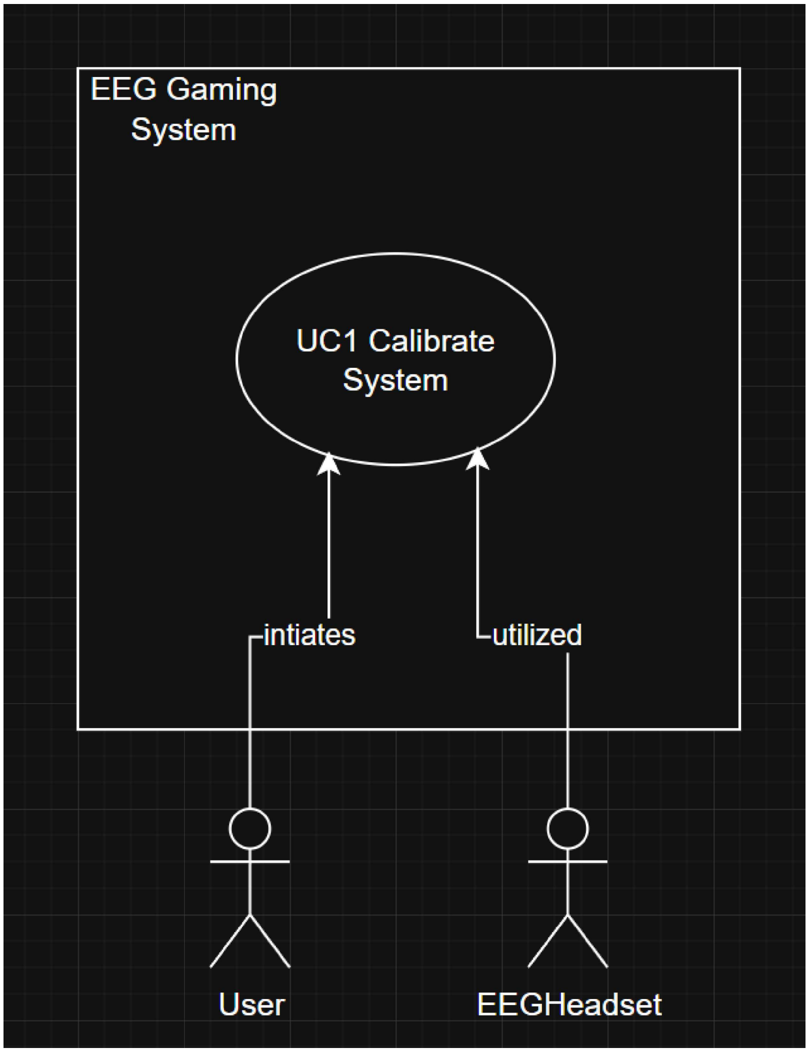

Figure 2.2: Use Case Diagram for UC1 – Calibrate System

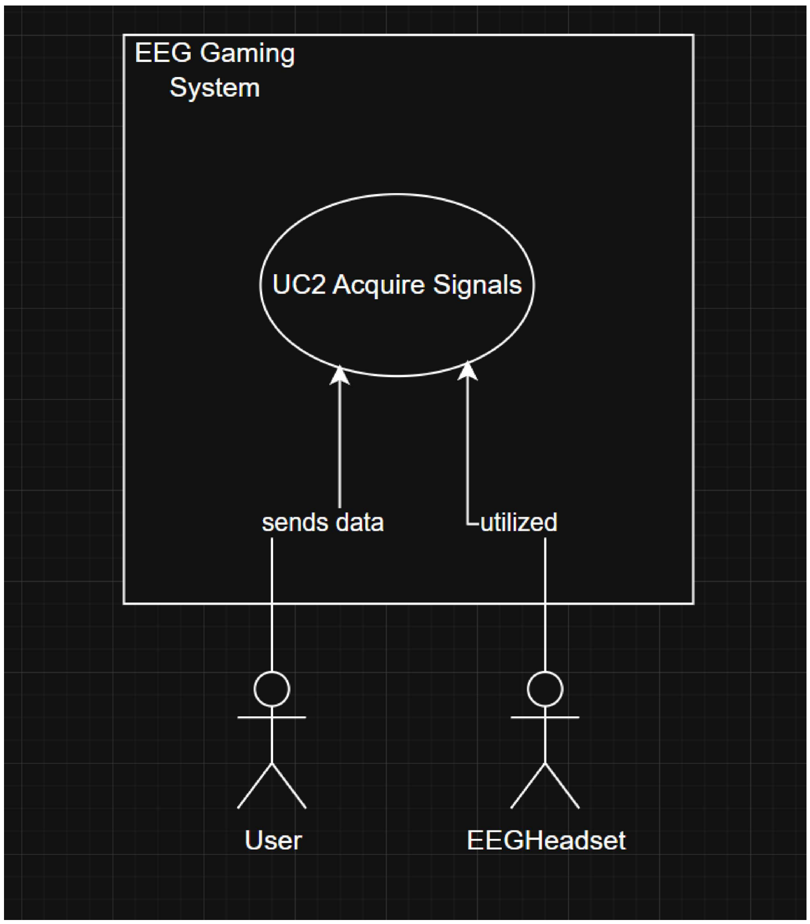

Figure 2.3: Use Case Diagram for UC2 – Acquire Signals

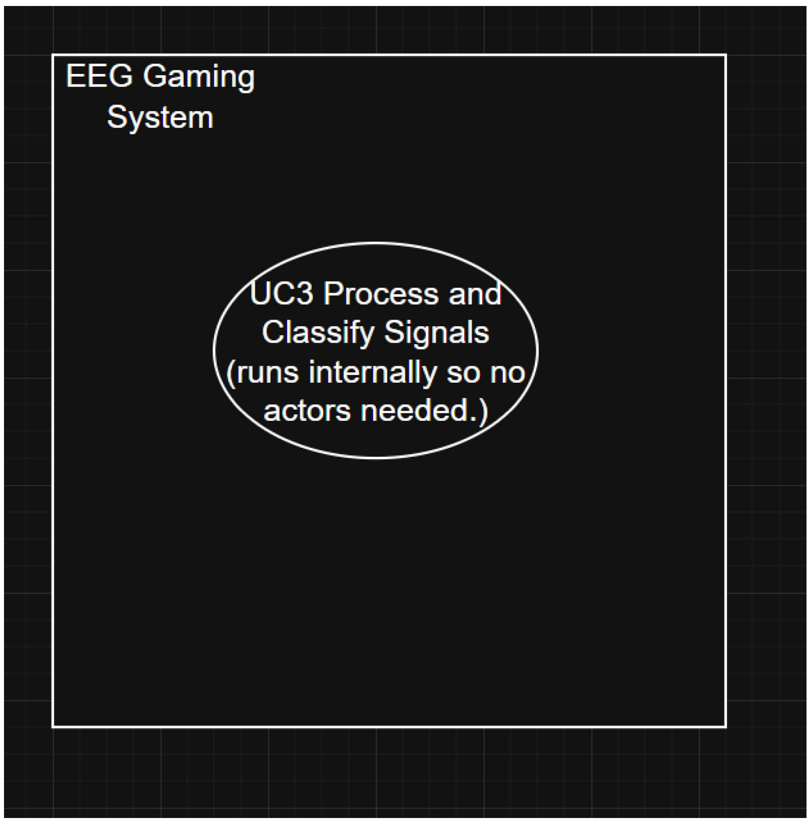

Figure 2.4: Use Case Diagram for UC3 – Process and Classify Signals

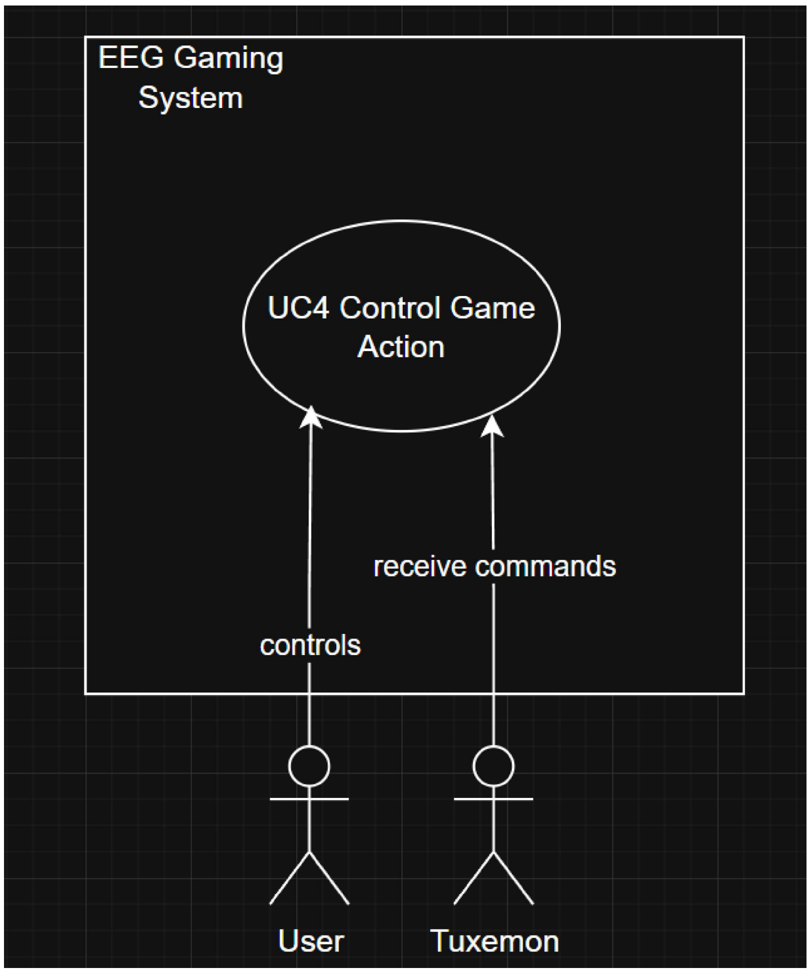

Figure 2.5: Use Case Diagram for UC4 – Control Game Actions

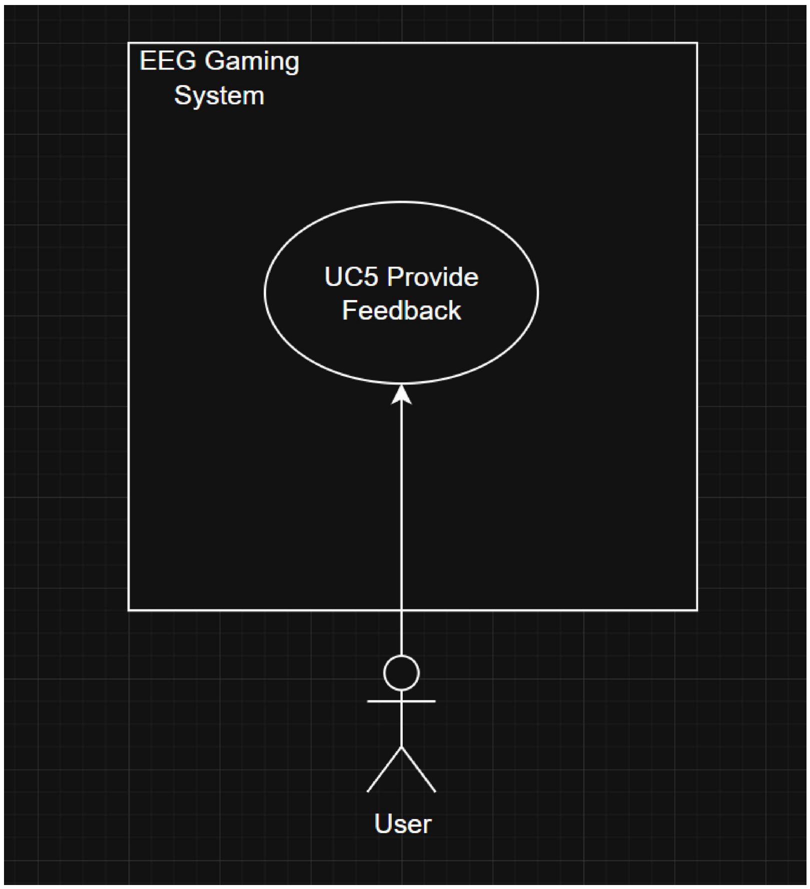

Figure 2.6: Use Case Diagram for UC5 – Provide Feedback

2.2 Functional Requirements

The functional requirements are categorized by modules as follows:

2.2.1 Module 1: Signal Acquisition

1. **FR1.1:** The system shall connect to the EEG headset and stream brain signals in real time.
2. **FR1.2:** The system shall capture gyroscope (IMU) data concurrently.
3. **FR1.3:** Data streams shall be synchronized with timestamps.
4. **FR1.4:** The sampling rate shall not be lower than 128 Hz.

2.2.2 Module 2: Signal Processing and Classification

1. **FR2.1:** Apply filtering and artifact removal to incoming EEG data.
2. **FR2.2:** Detect blinks via amplitude thresholding or ICA.
3. **FR2.3:** Use EEGNet to classify EEG data into Rest, Hand Imagery, or Blink.
4. **FR2.4:** Integrate gyro features with EEG features for enhanced accuracy.
5. **FR2.5:** Update model parameters during calibration for user adaptation.

2.2.3 Module 3: Game Integration

1. **FR3.1:** Map EEG and gyro outputs to keyboard or mouse events.
2. **FR3.2:** Ensure game response latency <= 500 ms.
3. **FR3.3:** Provide an option to configure control mappings for different games.

2.2.4 Module 4: Evaluation and Feedback

1. **FR4.1:** Display live confidence levels and signal quality.
2. **FR4.2:** Log session accuracy, latency, and user activity.
3. **FR4.3:** Generate post-session performance reports.

2.3 Non-Functional Requirements

2.3.1 Reliability

- **REL-1:** The system shall maintain >= 90% uptime during operation.
- **REL-2:** Auto-reconnect to headset within 5 seconds of signal loss.

2.3.2 Usability

- **USE-1:** Calibration shall complete within 15 minutes.
- **USE-2:** Visual indicators for feedback shall be intuitive.
- **USE-3:** Interface shall be learnable within 20 minutes of use.

2.3.3 Performance

- **PER-1:** EEGNet inference time shall not exceed 500 ms per window.
- **PER-2:** Maintain average classification accuracy >= 75%.
- **PER-3:** System shall process and display EEG data at >= 20 FPS.

2.3.4 Security

- **SEC-1:** All EEG and gyro data shall be stored locally.
- **SEC-2:** Only authorized users may access recorded sessions.
- **SEC-3:** System shall anonymize all user data in logs.

2.4 Domain Model

The domain model represents the conceptual structure of MindPlay, showing how its components interact to form a functional BCI system.

- **User:** Provides EEG and gyro input through mental imagery and movement.
- **EEG Device:** Captures brain activity in real time.
- **Signal Processor:** Filters, cleans, and segments input data.
- **Classifier:** Predicts intended actions using EEGNet.
- **Game Interface:** Maps classified actions to actual gameplay controls.
- **Game:** External application receiving control inputs.

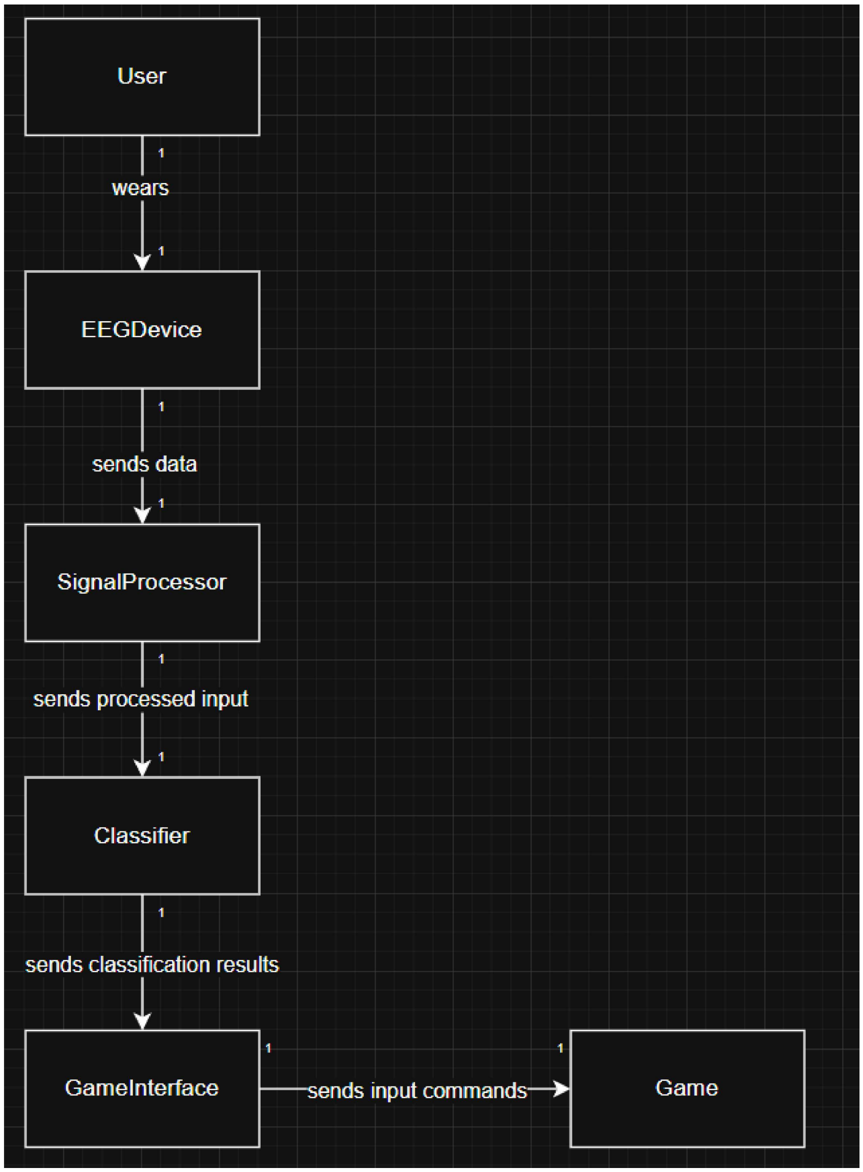

Figure 2.7: Domain Model for EEG-Based Game Control System (MindPlay)

Chapter 3

System Overview

This chapter provides a comprehensive description of the system's functionality, architectural context, and design principles. The MindPlay system represents an integrated Brain-Computer Interface (BCI) solution that transforms neural signals and motion data into actionable game commands, creating an accessible gaming experience for users with motor impairments.

3.1 Architectural Design

The MindPlay system follows a **Sequential Pipeline Architecture**, which is ideal for real-time data processing applications. Data flows unidirectionally through a series of processing stages, with each stage transforming the input before passing it to the next. This architectural approach ensures modularity, making each component independently developable, testable, and maintainable.

The entire system is built upon the **Lab Streaming Layer (LSL)** framework, which serves as the central communication backbone, ensuring perfect synchronization of all data streams while maintaining low-latency performance essential for real-time gaming applications.

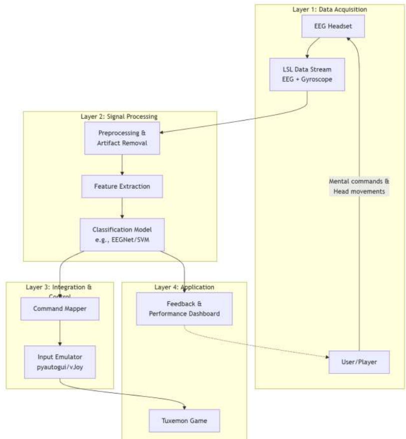

Figure 3.1: High-Level System Architecture showing the sequential pipeline from data acquisition to game control

Architecture Layer Explanation:

- **Layer 1: Data Acquisition** - The user interacts with the EEG headset, which produces continuous, synchronized LSL streams containing both EEG and gyroscope data.
- **Layer 2: Signal Processing & AI** - Raw data undergoes processing: initial cleaning and artifact removal, followed by feature extraction, culminating in classification models interpreting features into command intents (e.g., “Move Left”).
- **Layer 3: Integration & Control** - Classified intents are translated into specific keyboard or joystick commands, which are then emulated at the operating system

level.

- **Layer 4: Application** - The Tuxemon game receives and responds to these inputs, completing the control loop while a feedback dashboard provides users with real-time performance information.

3.2 Design Models

This section presents the design models applicable to our system, with detailed descriptions and visible diagrams that comprehensively represent the system's structure and behavior.

3.2.1 Design Models for Procedural Approach

Given the data-centric nature of our BCI pipeline and signal processing workflow, the system employs a **Procedural Development Approach**. The following models have been developed:

- **Data Flow Diagram** - Illustrates the transformation of data as it moves through system processes, extended to Level 2 to show detailed processing stages
- **System-level Sequence Diagram** - Captures the interaction sequence between major system components during real-time operation
- **Activity Diagram** - Models the workflow of user activities and system processes during calibration and gameplay
- **State Transition Diagram** - Represents the system's behavioral states and transitions in response to external events
- **Context Diagram (Level 0)** - The Level 0 Context Diagram provides a high-level overview of the MindPlay system, showing how it interacts with external entities such as the User, EEG Headset, and Game. It defines the system boundary and the major data flows between the MindPlay system and these external actors.
- **Context Diagram (Level 1)** - The Level 1 Context Diagram decomposes the MindPlay system into its major functional components, detailing how each subsystem (such as Signal Acquisition, Signal Processing, and Game Control Interface) interacts internally and with external entities.

- **Activity Diagram** - The Activity Diagram illustrates the dynamic behavior of the MindPlay system, showing the flow of control from initialization and calibration to classification and game control, including feedback and error handling.

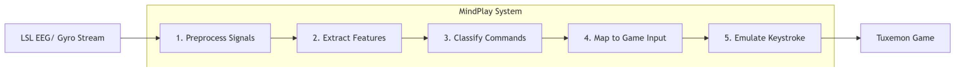

Figure 3.2: Level 1 Data Flow Diagram illustrating internal processes and data movement between system components

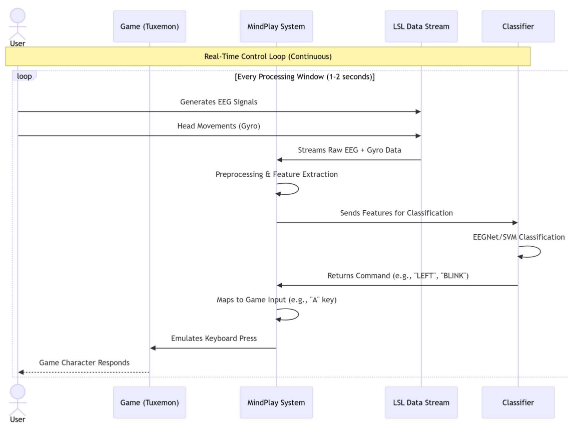

Figure 3.3: System Sequence Diagram showing real-time control loop interactions between user, system components, and game environment

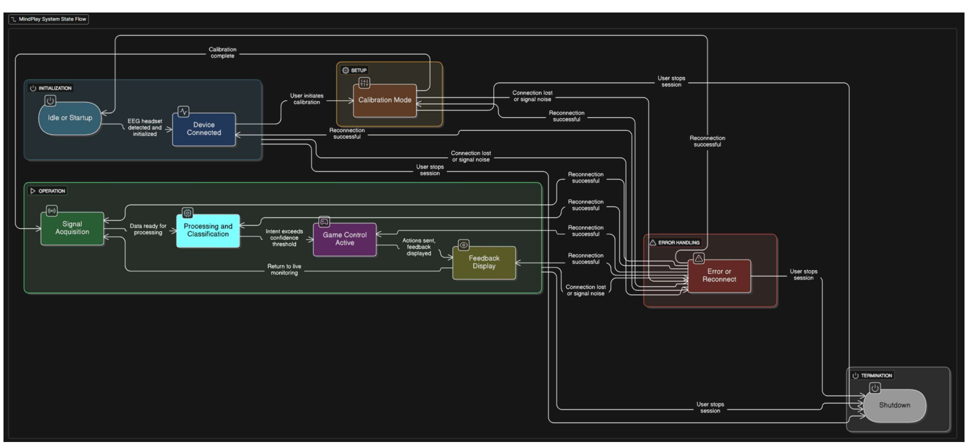

Figure 3.4: State Transition Diagram for EEG-Based Game Control System (MindPlay)

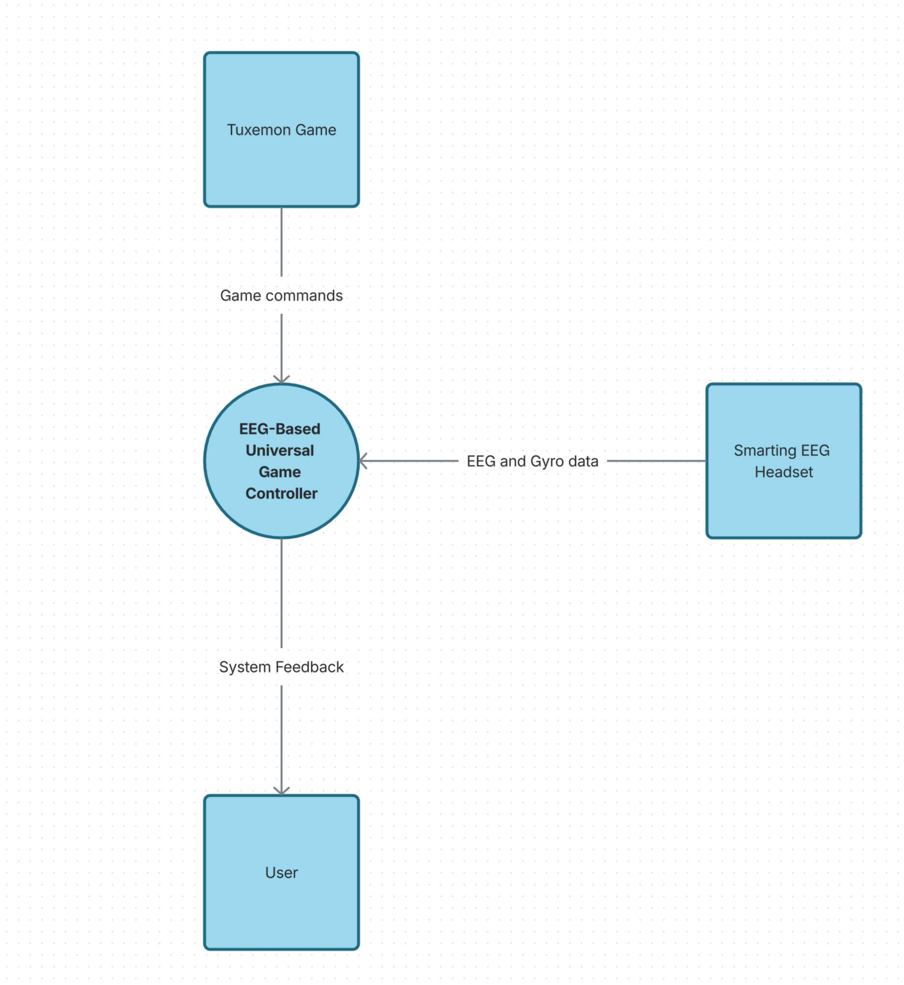

Figure 3.5: Context Diagram (Level 0) of MindPlay System

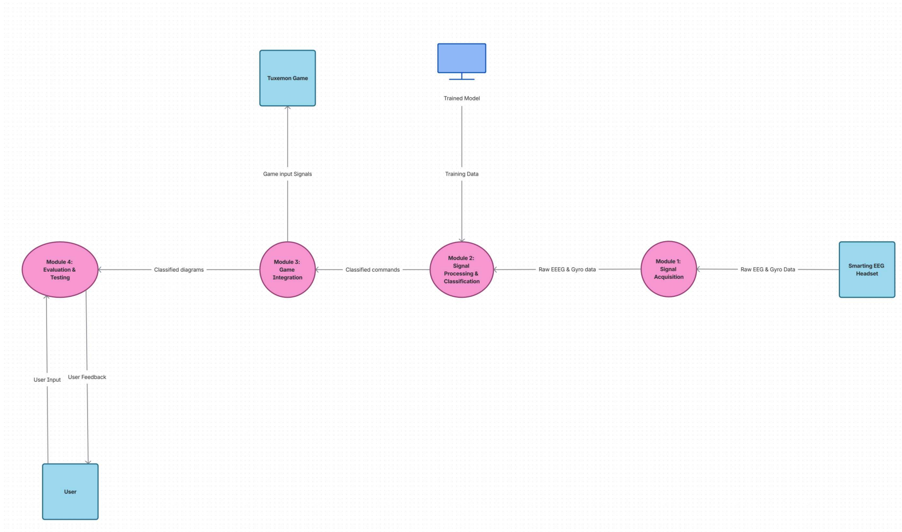

Figure 3.6: Context Diagram (Level 1) of MindPlay System

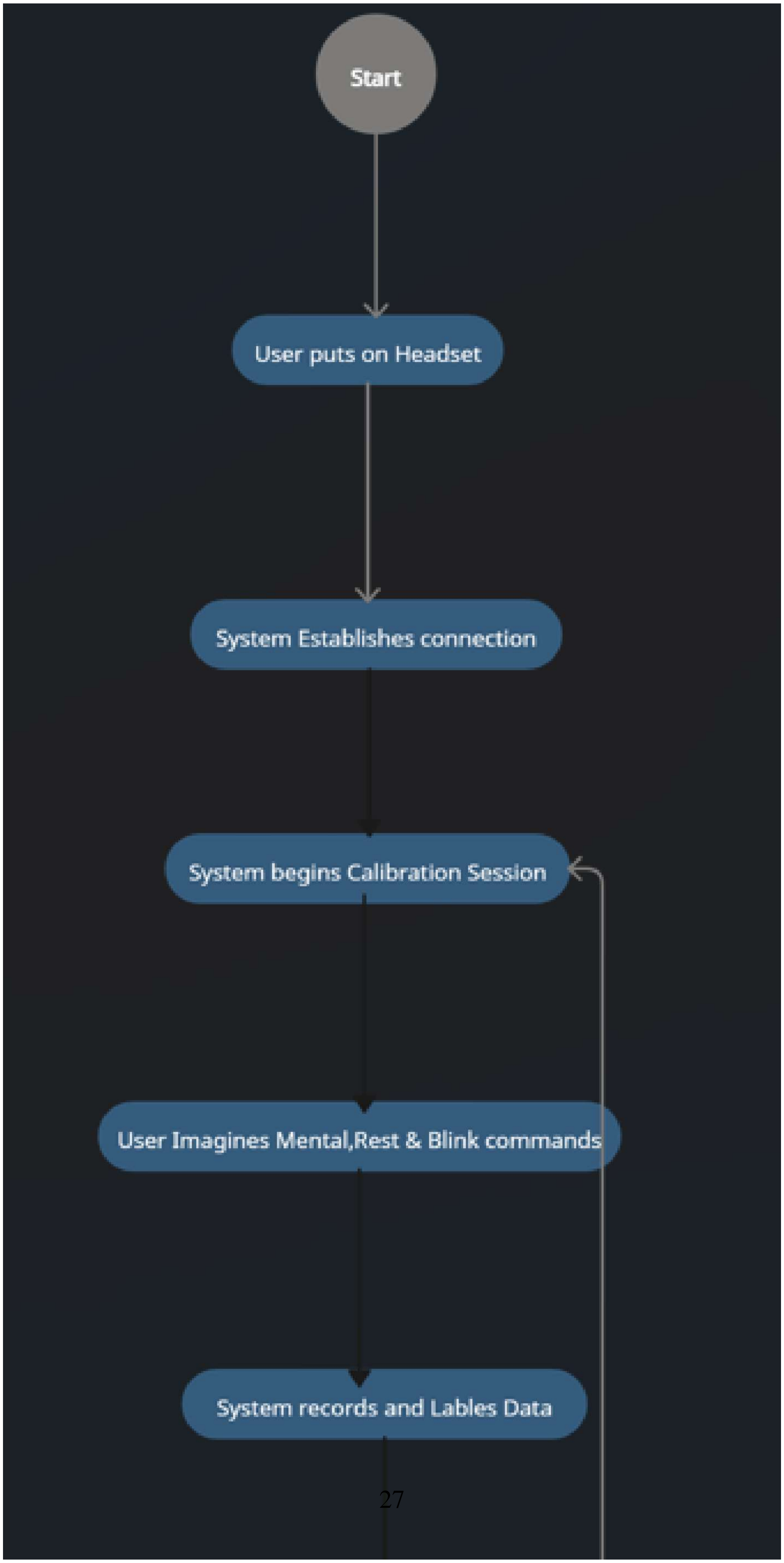

Figure 3.7: Activity Diagram for EEG-Based Game Control System (MindPlay) 1 of 2

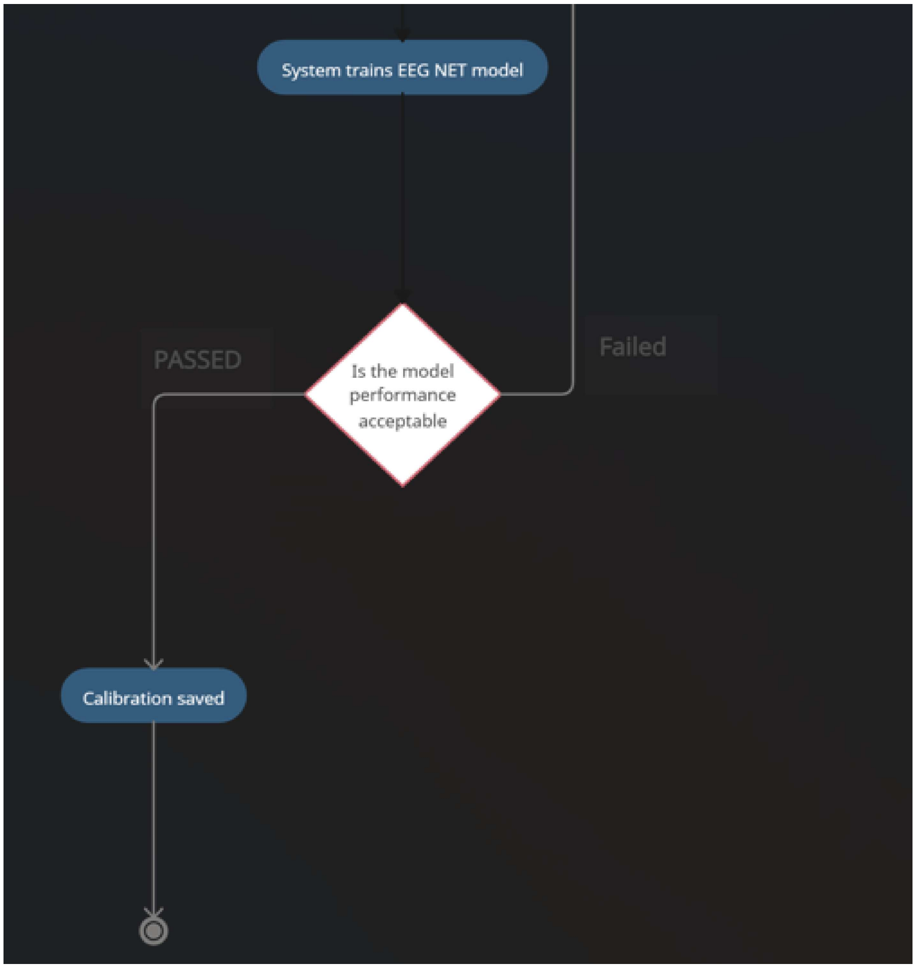

Figure 3.8: Activity Diagram for EEG-Based Game Control System (MindPlay) 2 of 2

3.3 Data Design

The information domain of the MindPlay system centers around time-series physiological data and its transformation into control commands. This section explains how major data entities are structured, processed, and organized throughout the system.

3.3.1 Data Structures and Processing

- **Raw Data Streams** - EEG data is structured as multi-channel arrays (24 channels × N samples) with microvolt measurements, while gyroscope data is maintained as 3-axis vectors (X, Y, Z), both synchronized via LSL timestamps

- **Processed Data** - Filtered epochs are maintained as overlapping time windows (1-2 seconds) in memory, with feature vectors stored as numerical arrays for immediate classification

- **Command Data** - Classified outputs are structured as intent labels with confidence scores for mapping to game actions

3.3.2 Data Storage and Organization

- **Model Storage** - Pre-trained EEGNet classifier parameters persisted as '.h5' or '.pkl' files on local disk
- **Configuration Data** - User preferences and key mappings stored in JSON configuration files
- **Session Logs** - Performance metrics and anonymized usage data recorded in structured CSV files for evaluation purposes
- **Calibration Data** - Temporary in-memory storage during user calibration sessions, with no persistent storage to ensure privacy

3.3.3 Data Flow Architecture

The system employs a hybrid storage strategy: **in-memory buffers** for real-time processing performance during active sessions, and **local file system storage** for configuration, models, and anonymized analytics. All sensitive user data is processed transiently with no persistent storage, adhering to our security requirements.

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